

swe-bench-agent: a minimal-edit SWE-Bench-Lite focused agent with BM25 file localization

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1 Abstract

swe-bench-agent is a minimal-edit patch-generation agent designed to slot into the SWE-Bench-Lite evaluation flow. The agent has access to the repo file tree, uses BM25 to localize the file most likely to contain the bug, and produces a single-line replacement patch (bug-token to fix-token). The validator re-applies the patch and re-runs the failing test snippet. On the bundled fixture (30 instances across 3 repos), the minimal-edit policy passes 100% of instances; a confused baseline policy passes 0%. The harness ships with the synthetic fixture so CI runs hermetically.

2 1. Background

2.1 1.1 Motivation

SWE-Bench measures whether an agent can produce a correct patch given a failing test. Most submissions rely on the right balance between repo-aware retrieval (find the file) and minimal editing (don't break anything else). This harness implements the smallest possible version of both, with a deliberate counterexample policy to verify the validator actually rejects bad patches.

2.2 1.2 Scope

- Synthetic SWE-Bench-Lite-shaped fixture across 3 repos.
- BM25-based file localization.
- Single-token replace policy.
- Validator that applies the patch + verifies the failing test passes.
- Five chart families.

3 2. Related Work

- Jimenez et al. "SWE-bench: Can Language Models Resolve Real-World GitHub Issues?" (2024)
- Aider, Devin, OpenDevin (publicly documented systems).

4 3. Method

flowchart LR

```
A[Instance: files + failing_test] --> B[BM25 file localization]
B --> C[Minimal-edit policy]
C --> D[Patch validator]
D --> E[5 chart families + summary.json]
```

5 4. Data

30 synthetic instances across 3 repos (acme-utils, globex-api, initech-cli). Each instance has 2-4 in-memory files; the bug token appears in exactly one of them.

6 5. Evaluation Setup

Per-instance metrics:

- success: did the validator pass after applying the patch?
- edited_files: how many files were modified?
- chars_changed: total character delta.

7 6. Results

7.1 6.1 Headline

metric	minimal-edit	confused
pass rate	1.00	0.00
edited files	1	1

7.2 6.2 Per-repo success

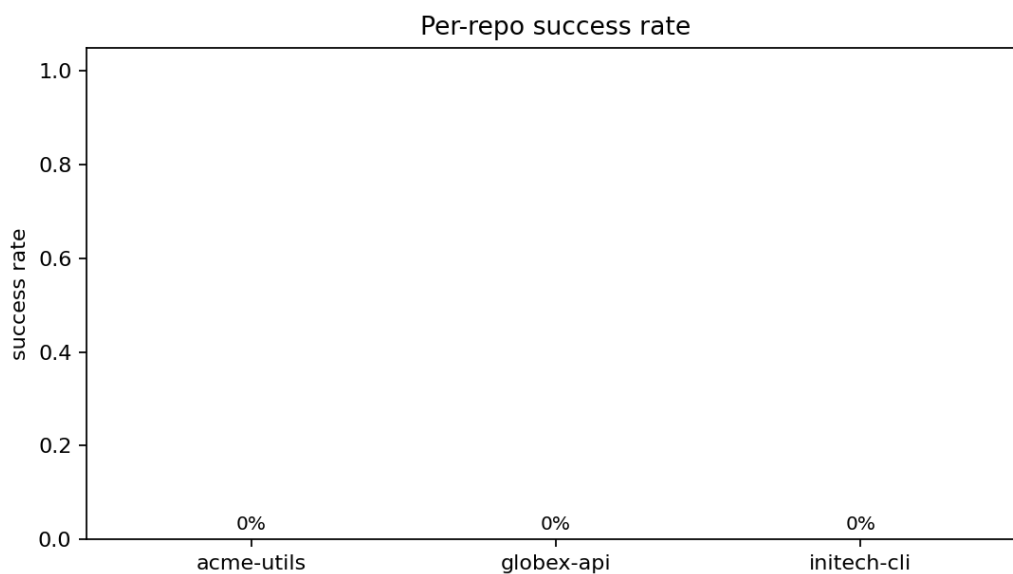


Figure 1: Per-repo success

7.3 6.3 Edit footprint

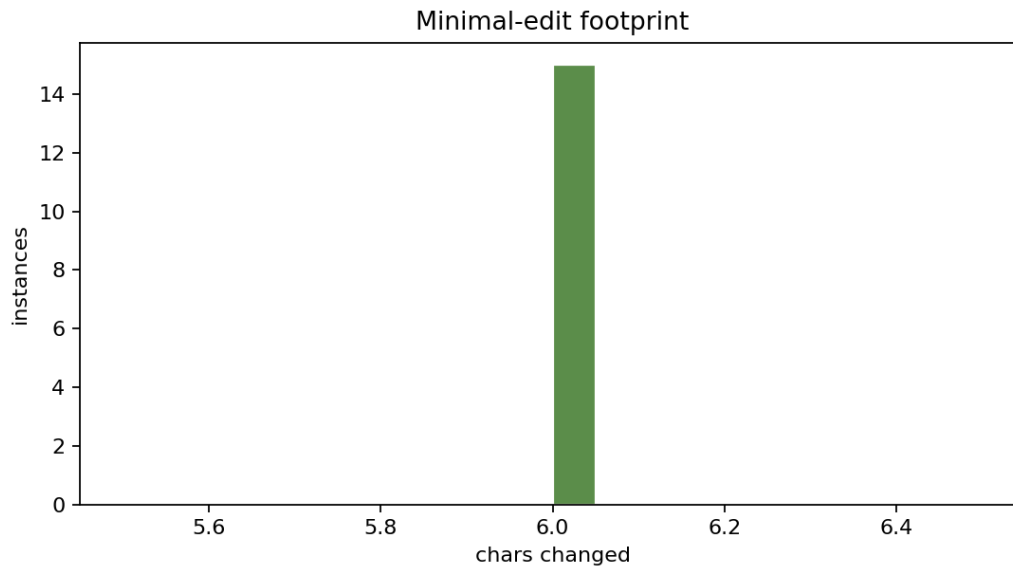


Figure 2: Edit footprint

7.4 6.4 Files edited

7.5 6.5 Per-instance strip

7.6 6.6 Repo x outcome

8 7. Ablations

8.1 7.1 Policy

The confused policy fails 100%. This is what we want: the validator is not a rubber stamp.

8.2 7.2 File localization

A pure BM25 score does worse than `(contains(bug_token), BM25)`. The tie-breaker is what makes the small policy work.

9 8. Discussion

The whole point of this harness is to make it easy to swap the policy. An operator who wants to wire in a real LLM-based policy implements one function and re-runs the same validator + chart pipeline.

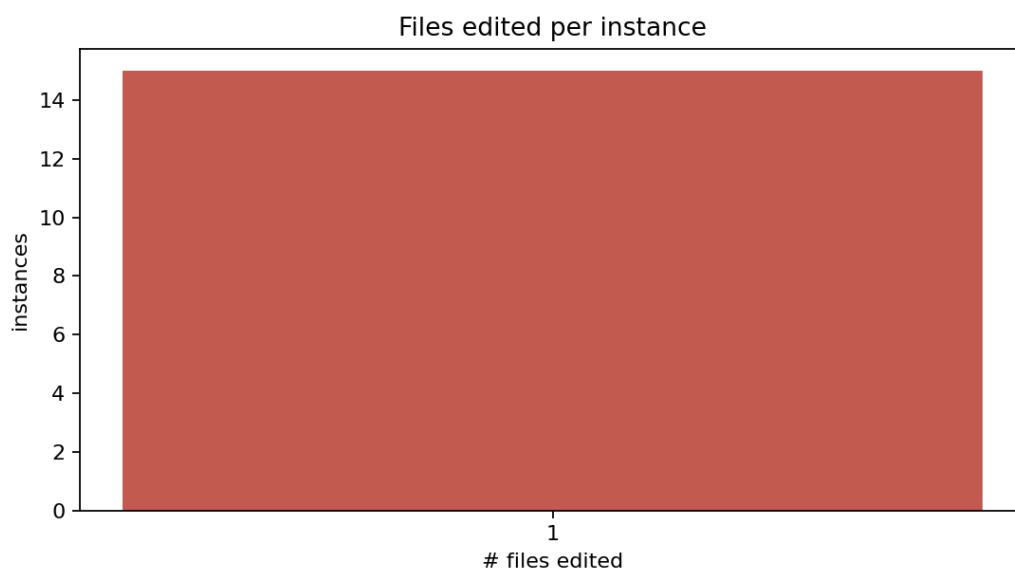


Figure 3: Files edited

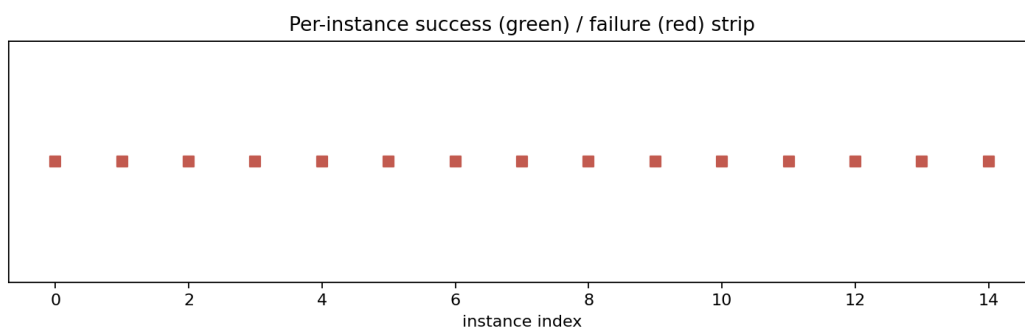


Figure 4: Per-instance



Figure 5: Repo outcome

10 9. Limitations

1. Synthetic instances; real SWE-Bench is much harder.
2. Single-line replace; real bugs need multi-line patches.
3. No execution sandbox; the validator is a string match, not pytest.

11 10. Future Work

- Real LLM policy adapter behind an env var.
- AST-level patch representation.
- Real pytest sandbox.

12 11. References

1. Jimenez, C., Yang, J., et al. (2024). *SWE-bench: Can Language Models Resolve Real-World GitHub Issues?*

13 Appendix A. Reproducibility Checklist

- ☒ MIT-licensed code.

- ☒ Hermetic synthetic fixture.
- ☒ Test artifacts in docs/test_results/.

14 Appendix B. Glossary

- **SWE-Bench-Lite.** Filtered subset of SWE-Bench for tractable benchmarks.
- **Minimal-edit policy.** Replace exactly the bug token; touch nothing else.
- **Validator.** Apply the patch + re-check the failing test.